

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [112, 102, 96, 86, 85, 78, 69, 55, 61, 72]

Tap amplitudes (pixels): [144.62710672622887, 132.13629327327143, 130.59862173851607, 134.6291201783626, 132.90974381135493, 138.40520221436765, 128.96898852049665, 114.28473213863697, 88.40814442120137, 118.15244390193544, 122.88612614937456]

GPT-5

MDS-UPDRS 3.4 Finger Tapping Scoring Analysis

Given data (10-tap sequence inferred from 11 amplitudes and 10 inter-tap periods):

- Tap count: 10 consecutive intervals (periods), 11 amplitudes
- Periods (frames): [112, 102, 96, 86, 85, 78, 69, 55, 61, 72]
- Amplitudes (pixels): [144.63, 132.14, 130.60, 134.63, 132.91, 138.41, 128.97, 114.28, 88.41, 118.15, 122.89]

Speed (periods)

- Trend is predominantly decreasing (faster tapping) from 112 → 55 frames over the first 8 intervals, then a slight increase to 61 and 72.
- No evidence of marked slowing; if anything, speed increases and then stabilizes slightly slower at the end. Variability is present but not indicative of moderate slowing.

Interpretation: No mild/moderate slowing pattern; no clear interruptions/arrests from period spikes.

Interruptions/hesitations

- Interruptions/freezes would typically appear as marked outlier period increases or missing taps. The sequence is continuous with no large spikes suggesting arrests.

- No more than occasional small variability; no evidence of 3–5 interruptions or >5 interruptions.

Interpretation: No clear interruptions/hesitations pattern that would elevate above slight at most.

Amplitude and decrement

- Amplitudes by tap (approx):
 1: 144.6
 2: 132.1
 3: 130.6
 4: 134.6
 5: 132.9
 6: 138.4
 7: 129.0
 8: 114.3
 9: 88.4
 10: 118.2
 11: 122.9
- There is a clear downward drift beginning around tap 7–9 (not immediately after the 1st tap). The amplitude falls substantially by tap 9 (to ~88), then partially recovers by taps 10–11.
- Pattern: amplitude decrement begins midway-to-late in the sequence, not just “near the end” (only last 1–2 taps), and not “starting after the 1st tap.”

Interpretation:

- “Slight” (1c): decrement near the end → would typically be limited to last taps only.
- “Mild” (2c): decrement midway through the 10-tap sequence → matches observed onset around taps 7–9, i.e., clearly before only the final tap or two, with a notable drop.

Overall best-fit criterion

- No clear interruptions/freezes.
- No mild or moderate slowing.
- Amplitude decrement emerges before the very end (midway/later), consistent with score 2c.

Final score: 2
